Lecture Notes on

Introduction to Astrophysics

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These notes summarize the first Astrophysics Club meeting. The goal is to outline the purpose of the club and introduce fundamental concepts about the scale and structure of the universe.

Contents

1 Purpose of the Club

The Astrophysics Club was created to:

- Explore astrophysics topics not typically covered in Fremont courses.
- Organize telescope observations and guest lectures.
- Discuss the physics governing the universe.
- Encourage critical thinking through competitions and projects.
- Build curiosity about large-scale structure and cosmic phenomena.

Leadership:

- President: Abir Mehta
- Vice President: Dhruv Lagu

2 Astrophysics Competitions

For students interested in deeper challenge:

- **IAAC** — International Astronomy and Astrophysics Competition.
- **USAAAO** — U.S. Astronomy and Astrophysics Olympiad.
- **USAAO** — U.S. Astronomy Olympiad.

Competitions emphasize:

- Problem solving
- Conceptual understanding
- Quantitative reasoning
- Application of physics to astronomy

3 Core Learning Objectives

Topics we aim to study include:

- Scale of the universe
- Celestial motion
- Light and electromagnetic radiation
- Stars and stellar evolution
- Galaxies

- Cosmology
- Exoplanets
- Relativity
- Black holes
- Telescopes and instrumentation
- History of astrophysics
- Current research frontiers

Underlying theme:

- Everything in astrophysics ultimately connects back to physics.
- Gravity, electromagnetism, thermodynamics, and nuclear physics govern cosmic behavior.

4 Scale of the Universe

Astrophysics is fundamentally about scale.

Key question:

- How large is the universe?
- How far apart are objects?
- How long do cosmic processes take?

Understanding scale is essential before studying advanced topics.

5 Distance Units in Astronomy

5.1 Light-Year

- A light-year is the distance light travels in one year.
- Speed of light:
$$c = 299{,}792{,}458 \text{ m/s}.$$
- Since light travels extremely fast, a light-year measures enormous distances.

Important clarification:

- A light-year is a unit of *distance*, not time.

5.2 Astronomical Unit (AU)

- 1 AU is the average distance between Earth and the Sun.
- Approximately 93 million miles.
- Useful for describing distances inside planetary systems.

5.3 Parsec

- 1 parsec $\approx$ 3.26 light-years.
- Common in professional astronomy.
- Derived from parallax measurements.

6 Cosmic Structure Hierarchy

From smallest to largest:

- Moon
- Planet
- Star
- Planetary system (e.g., Solar System)
- Star cluster
- Galaxy
- Galaxy group
- Galaxy cluster
- Supercluster
- Observable universe

6.1 Example: The Milky Way

- Diameter $\approx$ 100,000 light-years.
- Contains hundreds of billions of stars.
- Our Sun is located in one of its spiral arms.

7 Why Perspective Matters

Human intuition is built for everyday scales:

- meters
- kilometers
- seconds

Astrophysics operates on:

- billions of kilometers

- thousands of light-years
- billions of years

Developing intuition for these scales is the first step toward understanding astrophysical phenomena.

8 Looking Ahead

Future meetings will explore:

- Celestial motion
- Black holes
- Stellar evolution
- Relativity
- Cosmology

Astrophysics is only fun with people you know — invite friends and build the community.